

Deep Generative Models

Lecture 12

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AI Masters

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Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Reverse ODE

Let $\tau = 1 - t$ ($d\tau = -dt$):

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1 - \tau)d\tau$$

Reverse SDE

There's a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ in the following form:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t)d\bar{\mathbf{w}}, \quad dt < 0$$

Sketch of the Proof

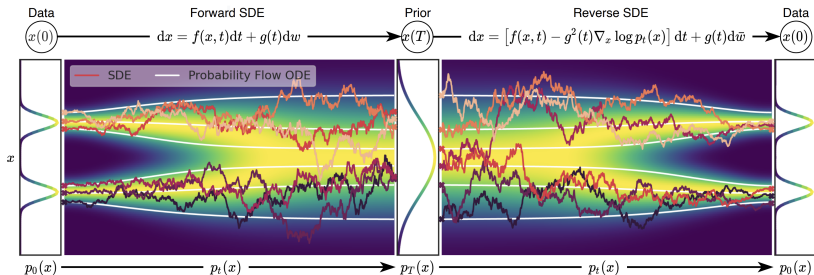
- ▶ Convert the initial SDE to the probability flow ODE.
- ▶ Reverse the probability flow ODE.
- ▶ Convert the reverse probability flow ODE to the reverse SDE.

Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \quad (\text{SDE})$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad (\text{probability flow ODE})$$

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t)d\bar{\mathbf{w}} \quad (\text{reverse SDE})$$



Recap of Previous Lecture

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0, 1]} \mathbb{E}_{q(\mathbf{x}(t) | \mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2$$

NCSN

$$q(\mathbf{x}(t) | \mathbf{x}(0)) = \mathcal{N}(\mathbf{x}(0), [\sigma^2(t) - \sigma^2(0)] \cdot \mathbf{I})$$

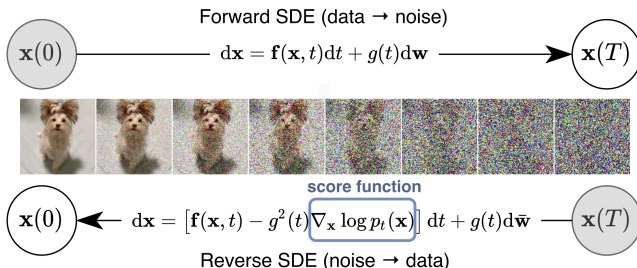
DDPM

$$q(\mathbf{x}(t) | \mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0) e^{-\frac{1}{2} \int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

Recap of Previous Lecture

Sampling

To sample, solve the reverse SDE using numerical solvers (SDEsolve).



- ▶ Discretizing the reverse SDE gives us ancestral sampling.
- ▶ If we use the probability flow ODE instead, then the reverse ODE yields DDIM sampling.

Recap of Previous Lecture

Consider ODE dynamics $\mathbf{x}(t)$ in the interval $t \in [0, 1]$ with $\mathbf{x}_0 \sim p_0(\mathbf{x}) = p(\mathbf{x})$, $\mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

Solving the continuity equation using the adjoint method is complicated and unstable.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Recap of Previous Lecture

Introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$
$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})).$$

- ▶ $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**
- ▶ $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t) \quad \Rightarrow \quad \frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, \mathbf{z}, t)$$

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

Recap of Previous Lecture

Flow Matching (FM)

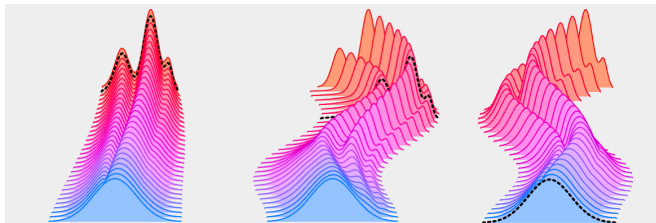
$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, \mathbf{z}, t)\|^2 \rightarrow \min_{\theta}$$

Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimum for CFM.



Tong A., et al. *Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport*, 2023

Outline

1. Conditional Flow Matching
2. One-Sided Conditioning
3. Two-Sided Conditioning

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1. Conditional Flow Matching

2. One-Sided Conditioning

3. Two-Sided Conditioning

Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ &= \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Proof

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ &= \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} [\|\mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] + \text{const}(\theta) \end{aligned}$$

Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

Proof

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} [\|\mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] + \text{const}(\theta) \\ \mathbb{E}_{p_t(\mathbf{x})} [\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)] &= \int p_t(\mathbf{x}) \left[\mathbf{v}_{\theta}^{\top}(\mathbf{x}, t) \left(\int p_t(\mathbf{z}|\mathbf{x}) \mathbf{v}(\mathbf{x}, \mathbf{z}, t) d\mathbf{z} \right) \right] d\mathbf{x} \end{aligned}$$

Conditional Flow Matching

Theorem

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Conditional Flow Matching

Theorem

$$\begin{aligned} \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 &= \\ = \arg \min_{\theta} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\theta}(\mathbf{x}, t)\|^2 \end{aligned}$$

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Conditional Flow Matching: Continuity Equations

Continuity Equations

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t) \cdot p_t(\mathbf{x}))$$
$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) \cdot p_t(\mathbf{x}|\mathbf{z}))$$

What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

Conditional Flow Matching: Continuity Equations

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What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.

Conditional Flow Matching: Continuity Equations

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What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

- ▶ $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- ▶ But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t)p_t(\mathbf{x})) = 0$ can generate a valid probability path.

Conditional Flow Matching: Continuity Equations

Continuity Equations

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- ▶ $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- ▶ But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t)p_t(\mathbf{x})) = 0$ can generate a valid probability path.
- ▶ Moreover, not every given path $p_t(\mathbf{x})$ can actually arise from particles moving under some velocity field $\mathbf{v}(\mathbf{x}, t)$.

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Conditional Flow Matching: Paths and Vector Fields

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Conditional Flow Matching: Paths and Vector Fields

Probability Paths

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Vector fields

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z}$$

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

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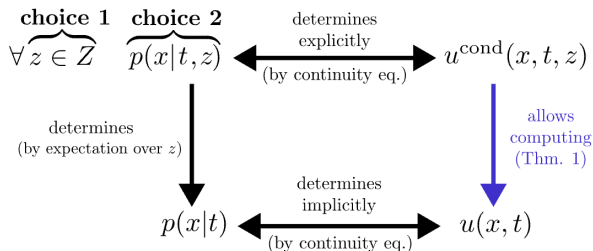
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Vector fields

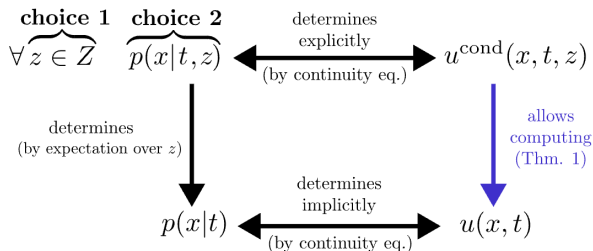
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- ▶ $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ and $p_t(\mathbf{x}|\mathbf{z})$ uniquely determine $\mathbf{v}(\mathbf{x}, t)$.
- ▶ But marginal velocity does not determine conditional velocities (many decompositions exist).

From Marginal to Conditional Paths

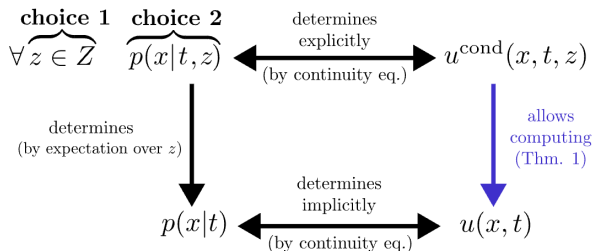


From Marginal to Conditional Paths



- ▶ We don't want to directly model $p_t(\mathbf{x})$, since it's complex.
- ▶ We've shown it's possible to solve the CFM task instead of the FM task.

From Marginal to Conditional Paths



- ▶ We don't want to directly model $p_t(\mathbf{x})$, since it's complex.
- ▶ We've shown it's possible to solve the CFM task instead of the FM task.
- ▶ Let's choose a convenient conditioning latent variable $\mathbf{z}$.
- ▶ We'll parametrize $p_t(\mathbf{x}|\mathbf{z})$ instead of $p_t(\mathbf{x})$. It should satisfy the following constraints:

$$p(\mathbf{x}) = \mathcal{N}(0, \mathbf{I}) = \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}); \quad p_{\text{data}}(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}).$$

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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Conditional Flow Matching: Training and Sampling

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Training

1. Sample a time $t \sim U[0, 1]$ and $\mathbf{z} \sim p(\mathbf{z})$.
2. Draw $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
3. Compute the loss $\mathcal{L} = \|\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

Gaussian Conditional Probability Paths

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Gaussian Conditional Probability Paths

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Open Questions

Q1 How should we choose the conditioning latent variable $\mathbf{z}$?

Q2 How can we define $p_t(\mathbf{x}|\mathbf{z})$ to enforce the constraints?

Gaussian Conditional Probability Paths

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Gaussian Conditional Probability Path [Q2]

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z}))$$

Gaussian Conditional Probability Paths

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Open Questions

Q1 How should we choose the conditioning latent variable $\mathbf{z}$?

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Gaussian Conditional Probability Path [Q2]

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mu_t(\mathbf{z}), \sigma_t^2(\mathbf{z}))$$

- ▶ There are infinitely many vector fields that generate a particular probability path.
- ▶ Let's consider the following dynamics:

$$\mathbf{x}_t = \mu_t(\mathbf{z}) + \sigma_t(\mathbf{z}) \odot \epsilon, \quad \text{with fixed } \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

Deriving the Conditional Vector Field

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})) ; \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

Is it possible to derive the expression for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$?

Deriving the Conditional Vector Field

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})) ; \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}$$

Is it possible to derive the expression for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$?

Statement

$$\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) = \boldsymbol{\mu}'_t(\mathbf{z}) + \frac{\boldsymbol{\sigma}'_t(\mathbf{z})}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

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Outline

1. Conditional Flow Matching
2. One-Sided Conditioning
3. Two-Sided Conditioning

Endpoint Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Let define our latent variable $\mathbf{z}$.

Endpoint Conditioning

Conditional Flow Matching

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Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = \mathbf{x}_1$. Then $p(\mathbf{z}) = p_1(\mathbf{x}_1)$.

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Conditional Flow Matching

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One-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

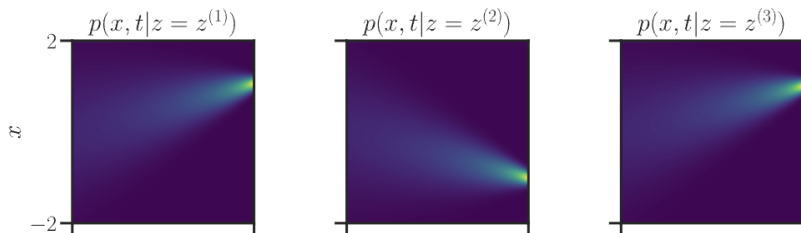
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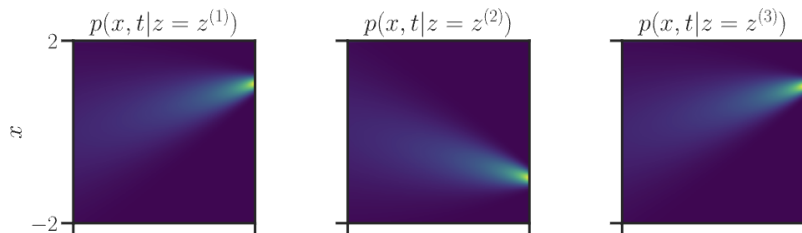


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Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases}$$

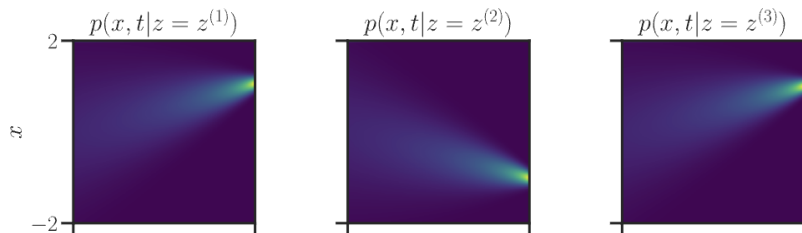
image credit: *A Visual Dive into Conditional Flow Matching*

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image credit: A Visual Dive into Conditional Flow Matching

One-Sided Conditioning: Conditional Vector Field

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$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \mu'_t(\mathbf{x}_1) + \frac{\sigma'_t(\mathbf{x}_1)}{\sigma_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \mu_t(\mathbf{x}_1))$$

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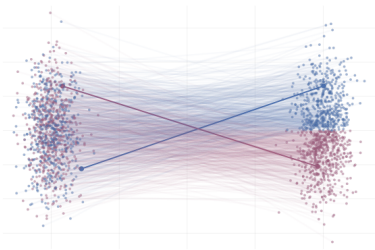
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The conditional vector field $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t)$ defines straight lines between $p_{\text{data}}(\mathbf{x})$ and $\mathcal{N}(0, \mathbf{I})$.

One-Sided Conditioning: CFM Objective

$$\begin{aligned} & \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_1)} \left\| \frac{\mathbf{x}_1 - \mathbf{x}}{1 - t} - \mathbf{v}_\theta(\mathbf{x}, t) \right\|^2 \end{aligned}$$

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One-Sided Conditioning: CFM Objective

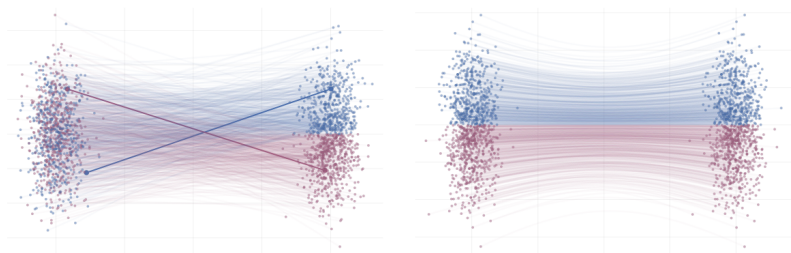
$$\begin{aligned} & \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_1)} \left\| \frac{\mathbf{x}_1 - \mathbf{x}}{1 - t} - \mathbf{v}_\theta(\mathbf{x}, t) \right\|^2 = \\ &= \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(t\mathbf{x}_1 + (1 - t)\mathbf{x}_0, t)\|^2 \end{aligned}$$

- ▶ We fit straight lines between the noise distribution $p(\mathbf{x})$ and the data distribution $p_{\text{data}}(\mathbf{x})$.
- ▶ The **marginal** path $p_t(\mathbf{x})$ does not give straight lines.

One-Sided Conditioning: CFM Objective

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CFM with One-Sided Conditioning: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\theta}(\mathbf{x}_t, t)\|^2 \rightarrow \min_{\theta}$$

Training

1. Sample $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$.
2. Sample time $t \sim U[0, 1]$ and $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
3. Obtain the noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
4. Compute the loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\theta}(\mathbf{x}_t, t)\|^2$.

CFM with One-Sided Conditioning: Training and Sampling

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Training

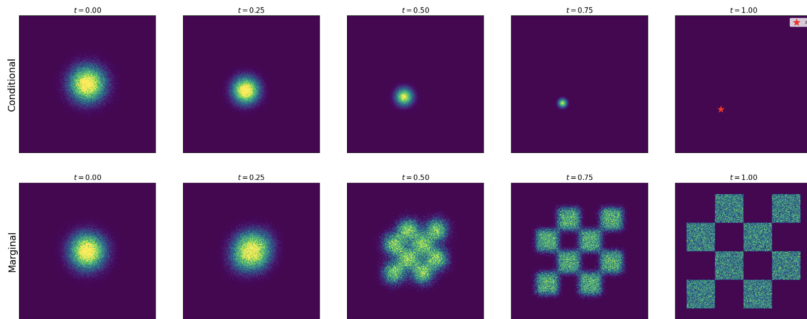
1. Sample $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$.
2. Sample time $t \sim U[0, 1]$ and $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
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Sampling

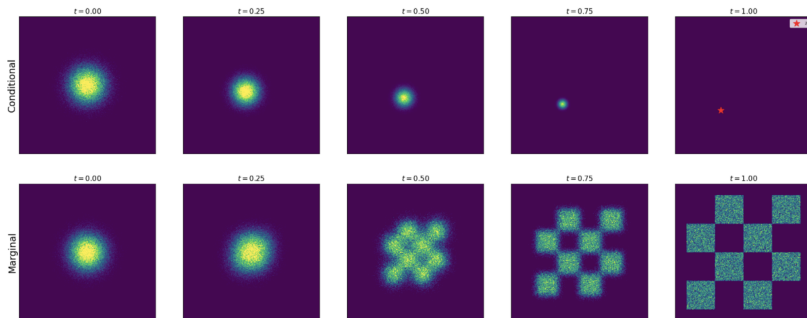
1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1)$$

Conditional vs. Marginal Paths



Conditional vs. Marginal Paths



- ▶ One-sided conditioning gives us the way to construct generative model.
- ▶ Now we extend it to image-to-image formulation (mapping between two distinct $p_{\text{data}_1}(\mathbf{x})$ and $p_{\text{data}_2}(\mathbf{x})$).

Outline

1. Conditional Flow Matching
2. One-Sided Conditioning
3. Two-Sided Conditioning

Pair Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$. Then $p(\mathbf{z}) = p(\mathbf{x}_0, \mathbf{x}_1) = p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) p_0(\mathbf{x}_0) p_1(\mathbf{x}_1) d\mathbf{x}_0 d\mathbf{x}_1$$

Pair Conditioning

Conditional Flow Matching

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We must enforce boundary constraints:

$$\begin{cases} p_0(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}); \\ p_1(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) \end{cases}$$

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Conditional Flow Matching

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Two-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

Two-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \\ \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) = \epsilon \end{cases}$$

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Two-Sided Conditioning

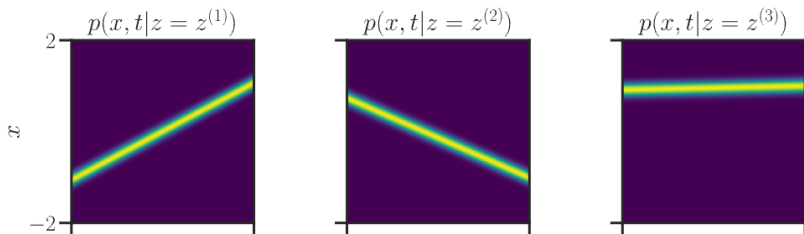
$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

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Flow Matching: One-Sided vs. Two-Sided Conditioning

$$\mathbf{z} = \mathbf{x}_1$$

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1, (1-t)^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

$$\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$$

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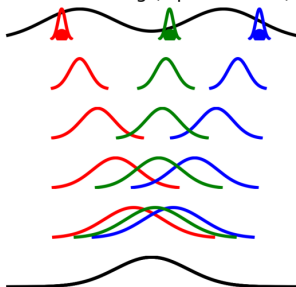
$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

$$\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$$

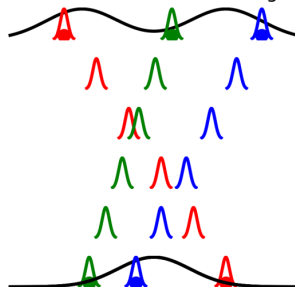
$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2\mathbf{I})$$

$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Flow Matching (Lipman et al.)



Conditional Flow Matching



Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2 \mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \mu'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\sigma'_t(\mathbf{x}_0, \mathbf{x}_1)}{\sigma_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \mu_t(\mathbf{x}_0, \mathbf{x}_1))$$

Two-Sided Conditioning: Vector Field and Objective

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Conditional Vector Field

$$\begin{aligned} \frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) &= \boldsymbol{\mu}'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_0, \mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1)) \\ \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) &= \mathbf{x}_1 - \mathbf{x}_0 \end{aligned}$$

Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N} \left(t\mathbf{x}_1 + (1 - t)\mathbf{x}_0, \epsilon^2 \mathbf{I} \right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$$

Conditional Vector Field

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Conditional Flow Matching

$$\begin{aligned} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 = \\ \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1)} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \end{aligned}$$

Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2 \mathbf{I}); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \boldsymbol{\mu}'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_0, \mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1))$$
$$\mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \mathbf{x}_1 - \mathbf{x}_0$$

Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 =$$
$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1)} \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2$$

- ▶ This yields the same procedure as for one-sided conditioning!
- ▶ Now, we do not require that $p_0(\mathbf{x})$ is necessarily $\mathcal{N}(0, \mathbf{I})$.

CFM with Two-Sided Conditioning: Training and Sampling

- ▶ This conditioning allows us to transport any distribution $p_0(\mathbf{x})$ to any distribution $p_1(\mathbf{x})$.
- ▶ It's possible to apply this approach to paired tasks, e.g., style transfer.

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Training Procedure

1. Sample $(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)$.
2. Sample time $t \sim U[0, 1]$.
3. Compute the noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
4. Compute the loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

CFM with Two-Sided Conditioning: Training and Sampling

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Training Procedure

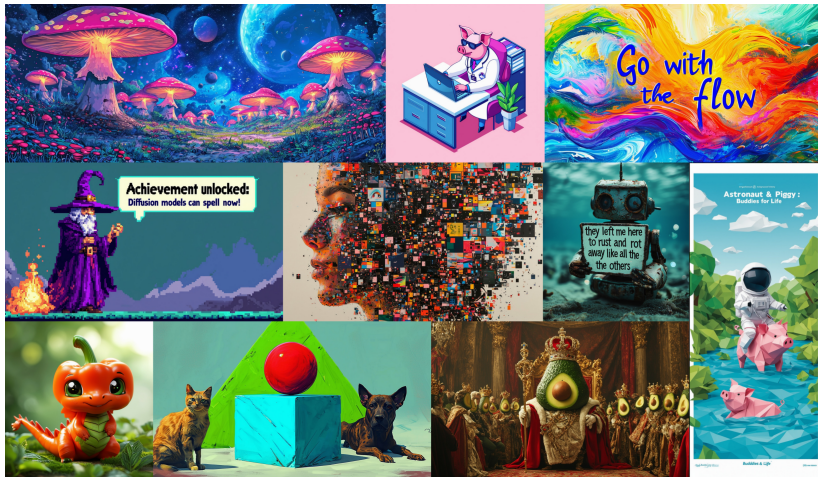
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4. Compute the loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_\theta(\mathbf{x}_t, t)\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim p_0(\mathbf{x})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1)$$

Stable Diffusion 3: Scalable Flow Matching



Esser P., et al. *Scaling Rectified Flow Transformers for High-Resolution Image Synthesis*, 2024

Summary

- ▶ Conditional flow matching makes the FM objective tractable.
- ▶ One-sided conditioning uses endpoint conditioning $\mathbf{z} = \mathbf{x}_1$ and serves as an effective FM technique.
- ▶ Two-sided conditioning uses pair conditioning $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$ and yields the same procedure, but is more general (suitable for paired tasks).